

The Macro Alibi: Subjective Risk Attribution in Analyst Scenarios

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Abstract

We study how analysts attribute subjective risk in scenario-based valuation reports. Using sell-side analysts' valuations and accompanying narratives, we document a misattribution bias in downside narratives: analysts overweight macro risk in bear scenarios relative to base or bull scenarios. However, conditional on the same subsequent market state, realized CAPM R^2 is similar across realized bear, base, and bull outcomes, implying that analysts overstate the systematic component of downside risk. Consistent with this interpretation, the bear–base macro-attention gap positively predicts subsequent analyst forecast errors, with stronger predictive power in nonlinear specifications. Portfolios formed on this bias-adjusted signal earn monthly CAPM alphas of up to 1.65%. Potential mechanisms include the availability heuristic and analysts' career concerns.

Keywords: Conditional CAPM, downside risk, analyst beliefs, large language model

JEL: C55, G11, G12, G14, G23

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1 Introduction

[name the bias: settle on a term, and modify intro, framework, empirical sections accordingly.

1. Subjective Conditional CAPM (SCCAPM)

- this is the framework through which we understand their bias. SCCAPM does not explain *what* the mistake is.
- our finding is that analysts *fail to condition*: they apply macro-dominant templates *unconditionally* across all market states. Calling the bias a SCCAPM implies analysts are running a conditional model (just subjectively). In other words, they should have conditioned, but didn't in their subjective model. This name may be a bit tautological.
- conditional CAPM literature mostly focuses on state-dependent *betas*, not R^2 or source of variation.

2. Macro Bear Bias (MBB)

- more accurate; catchy; less ambiguous
- MBB describes the observed pattern: bear narratives over-weight macro. We can provide accurate definition of the bias when first introduced; and use this 3-word shorthand afterwards.

3. Misattribution Bias: too generic

4. ...

] Analysts routinely summarize uncertainty through scenario-based valuation reports. In principle, these reports should do more than provide alternative price targets: they should also reveal how analysts think about the sources of upside and downside risk. A bear case may reflect exposure to aggregate macroeconomic shocks, or instead the possibility of a firm-specific collapse. This distinction matters because the two forms of risk have very different

implications for valuation, hedging, and expected return. Yet existing research has paid far more attention to the level of analysts' forecasts than to how analysts narrate the source of risk across scenarios. This paper studies that attribution problem directly. Using Morgan Stanley Risk-Reward reports, which require analysts to present bull, base, and bear price targets together with scenario-specific narratives, we ask a simple question: when analysts describe downside states, do they correctly identify whether that downside is systematic or idiosyncratic?

We show that the answer is no. Analysts systematically portray bear scenarios as disproportionately macro-driven relative to base and bull scenarios. We refer to this pattern as a subjective conditional CAPM: analysts behave as if downside states are especially exposed to market-wide risk, but they apply this intuition too broadly and too mechanically. The key point is not that analysts are wrong to recognize that downside comovement can be higher in bad times. Rather, the mistake is that they treat this state-dependent empirical regularity as a nearly unconditional template for writing bear-case narratives. In their reports, macro risk remains unusually prominent in bear scenarios across a wide range of future market environments.

To study this issue, we assemble a large sample of Morgan Stanley Risk-Reward reports from 2007 to 2025. These reports are especially well suited to our setting because they provide both structured scenario price targets and narrative explanations for each scenario. We use large language models to classify the narrative content of bull, base, and bear cases into macro-, industry-, and firm-level risk attribution. This design allows us to move beyond conventional analyst datasets that contain only point forecasts or target prices. Instead of observing merely what analysts predict, we can observe how they justify those predictions and how the justification changes when moving from the expected outcome to the downside tail.

Our first finding is a clear downside asymmetry in narrative attention. Bear-case narratives place substantially more weight on macro factors and less weight on firm-specific factors than either base- or bull-case narratives, while industry attention is comparatively similar

across scenarios. This asymmetry is remarkably stable over time. The pattern suggests that recession-style or aggregate-shock language functions as a default narrative frame for downside scenarios. In other words, analysts appear to rely on a macro-heavy bear template when articulating tail risk.

Our second finding is that this narrative asymmetry is not well matched to realized outcomes. If analysts are correct to emphasize macro risk in bear scenarios, then reports that ultimately realize a bear outcome should exhibit stronger ex post comovement with the market than reports that realize base or bull outcomes, conditional on the same subsequent market environment. We test this idea using realized CAPM R^2 over the year following report issuance. The data do not support the analysts' narrative weighting. Conditional on the same future market state, realized CAPM R^2 is similar across realized bear, base, and bull outcomes. What differs sharply is not realized systematic exposure, but analysts' ex ante description of that exposure. This gap implies that analysts overstate the systematic component of downside risk. The bias is therefore a miscalibration of source, not necessarily of magnitude: analysts may reasonably perceive downside risk as large, yet still misattribute that downside to macro forces when it is in fact more idiosyncratic.

We then examine whether this subjective risk attribution has asset-pricing implications. The central mechanism is that investors may absorb not only analysts' target prices but also the narrative framing that accompanies them. A bear narrative anchored in GDP contraction, higher rates, or tighter credit conditions encourages investors to interpret downside risk as contingent on a broad macro shock. In normal or strong market environments, such a shock appears remote, which makes the bear case seem less relevant and can lead investors to underweight firm-specific tail risk. Consistent with this mechanism, we find that the bear-base macro-attention gap predicts subsequent analyst forecast errors: firms whose downside narratives place unusually heavy emphasis on macro risk subsequently experience larger base-case forecast errors. Moreover, this predictive relation is highly nonlinear. Flexible nonlinear methods extract substantially more forecasting power than linear specifications, indicating that the misattribution of downside risk interacts with market

conditions in a complex way. Portfolios formed on this bias-adjusted signal earn economically meaningful abnormal returns.

Literature

Our paper connects to four strands of literature. [Ang and Chen \(2002\)](#) document that correlations between U.S. equities and the market are substantially higher for downside moves than for upside moves, and [Ang, Chen, and Xing \(2006\)](#) establish a downside beta premium in the cross-section of stock returns. [Lettau, Maggiori, and Weber \(2014\)](#) show that a conditional CAPM in which betas are elevated in bad states prices the cross-section of currency and equity returns, extending the partial rationality of state-dependent risk pricing to multiple asset classes. [Lewellen and Nagel \(2006\)](#) caution, however, that the conditional CAPM does not mechanically resolve standard asset-pricing anomalies, since beta variation would need to be implausibly large. [Lewellen, Nagel, and Shanken \(2010\)](#) further show that the standard cross-sectional testing approach is poorly suited to discriminating among conditional factor models. [Kacperczyk, Van Nieuwerburgh, and Veldkamp \(2016\)](#) and [Kacperczyk, Nieuwerburgh, and Veldkamp \(2014\)](#) show that fund managers optimally shift attention between aggregate market-timing and firm-specific stock-picking as a function of the business cycle: macro attention is concentrated in recessions, when systematic risk is most relevant, and dispersed in expansions. Our paper contrasts with this optimal attention-allocation result: analysts do not make the analogous shift in bear-case narrative construction. Bear narratives remain macro-heavy across all market regimes, suggesting that the narrative template is set at the time of scenario drafting without subsequent conditioning on realized states. These patterns form the genuine empirical foundation that analysts overgeneralize in bear-case scenario construction. We complement this literature by characterizing the limits of that generalization: the empirical regularity in downside comovement is state-dependent, and analysts apply macro-heavy bear narratives unconditionally even when the realized evidence calls for state-dependent conditioning. The macro bear bias is not a departure from the Ang-Chen fact—it is the failure to condition on it.

A second strand studies the economic narratives analysts construct to justify their forecasts—the qualitative reasoning that accompanies and shapes quantitative outputs.¹ Joos, Piotroski, and Srinivasan (2016) show that analyst-constructed risk scenarios contain information about fundamental uncertainty not captured by standard quantitative forecasts. Ke (2024) use analyst narratives to recover subjective return beliefs, documenting that stated reasoning carries information beyond point estimates. Bradshaw et al. (2025) analyze qualitative signals in analyst reports at scale, showing that textual content unlocks dimensions of information not reflected in price targets. Bastianello, H Décaire, and Guenzel (2024) show that forecasters with different internal models of the economy form persistently divergent expectations. We extend this literature by exploiting the *across-scenario* variation in narrative content—bear versus base versus bull—rather than the level of any single scenario. The three-scenario structure of the Morgan Stanley reports allows us to identify the incremental macro attribution that analysts add when moving from the expected outcome to the downside, which is precisely the miscalibration of interest and which a single-scenario analysis cannot detect.

Our paper also contributes to the emerging literature on narrative finance. Shiller (2019) argues that viral economic narratives—most potently, recession narratives—shape agents’ expectations and drive real outcomes. Our evidence complements this view: “recession risk” functions as a default bear-case template in analyst reports, applied regardless of whether the underlying downside risk is macro-driven. Addoum, Meursault, and Murfin (2024) document macro emphasis in debt investor narratives for corporate borrowers, providing complementary evidence from the bond side of the capital structure. We contrast with their level-based analysis by establishing that the equity bear narrative constitutes a state-

¹A complementary and burgeoning literature studies subjective beliefs embedded in analyst *numerical* forecasts of earnings and long-term growth. La Porta (1996) shows that stocks with high analyst long-term growth forecasts earn negative abnormal returns, an early indication that analyst growth expectations are systematically optimistic. Bordalo et al. (2019) and Bordalo et al. (2024) develop the diagnostic expectations model and show it accounts for the overreaction pattern in analyst earnings forecasts and the associated stock return anomalies. Bouchaud et al. (2019) document that analysts are slow to revise earnings expectations in response to profitability news, generating the profitability anomaly. De la O and Myers (2021) use analyst forecasts to measure subjective cash flow and discount rate expectations, finding that cash flow news drives return variation more than standard rational-expectations models imply.

dependent miscalibration: macro emphasis does not co-vary with the fraction of bear-case realizations that are genuinely systematic, which a level analysis cannot uncover.

Finally, our findings connect to the career concerns and herding literature in finance and economics. [Holmström \(1999\)](#) establishes the canonical model in which agents distort actions to signal ability, and [Scharfstein and Stein \(1990\)](#) show that career incentives lead agents to mimic each other even when private information would call for deviation. In the analyst context, [Hong and Kubik \(2003\)](#) show that analysts early in their careers issue more optimistic forecasts to manage the reputational cost of bold pessimistic calls, and [Chevalier and Ellison \(1999\)](#) document the same incentive structure for mutual fund managers. We reconcile this literature with the macro bear bias: attributing bear-case downside to macroeconomic forces shifts the perceived locus of the call from the analyst’s own judgment to an external macro shock, reducing reputational exposure if the forecast fails. This incentive to externalize blame complements the availability heuristic as a mechanism for why the macro bear bias is stable across firms, market regimes, and analyst cohorts.

The rest of the paper proceeds as follows. Section 2 develops the conceptual framework and introduces the misattribution bias. Section 3 describes the data, measurement, and empirical evidence on the misattribution bias. Section 4 examines the asset-pricing implications of this misattribution. Section 5 discusses potential mechanisms. Section 6 concludes.

2 Empirical Analysis

2.1 Data

Our paper uses data from the Risk-Reward Framework in Morgan Stanley analyst reports. In total, we collected 63,191 Risk-Reward reports from January 2007 to May 2025.

Traditional sell-side research often relied on single-point price targets or coarse recommendations (e.g., “buy,” “hold,” or “sell”), which can convey a false sense of precision and provide limited information about the range of possible future outcomes. To address this limitation, Morgan Stanley introduced the Risk-Reward Framework in 2007, requiring

analysts to present and justify 12-month stock-price expectations under three scenarios: bull, base, and bear. This framework was designed to provide a more informative and differentiated research product for institutional investors, who care not only about expected return but also about the distribution of potential outcomes and the sources of upside and downside risk. By explicitly presenting upside and downside scenarios, these reports help investors assess the skewness of expected payoffs, understand the assumptions underlying each case, and incorporate analyst views more directly into portfolio decisions. The reports are particularly valuable for our purposes because they provide both structured scenario-specific price targets and narrative discussion of the reasoning, key drivers, and investment debate behind those targets. For introductions to the framework, see [Weyns et al. \(2011\)](#) and [Srinivasan and Lane \(2011\)](#).

This dataset offers two main advantages. First, in addition to the base-case forecast, it contains analyst expectations under both bull and bear scenarios, which allows us to measure subjective upside and downside risk expectations separately. Second, the accompanying narratives explain the assumptions and reasoning behind each scenario. Using natural language processing (NLP) and large language models (LLMs), we can systematically extract and analyze these texts to identify the drivers of perceived upside and downside risks.

To ensure the validity of the extracted scenario price targets, we retain only reports in which the base-case target lies between the bull-case and bear-case targets.² This filter yields a final sample of 62,571 reports. Table 1 reports the number of available Risk-Reward reports and the number of covered firms by year.

2.2 Textual Identification of Subjective Risks

We focus on the narrative text in each Morgan Stanley Risk-Reward report to understand how analysts justify their bull, base, and bear case expectations. Specifically, we examine

²To construct the dataset, we use three rule-based extraction procedures tailored to changes in Morgan Stanley report layouts over 2007–2014, 2015–2016, and 2017–2026. The procedures identify the relevant report section and extract the bull, base, and bear price targets. The resulting outputs are consolidated into a structured dataset for empirical analysis using Python packages including PyMuPDF and pdfplumber.

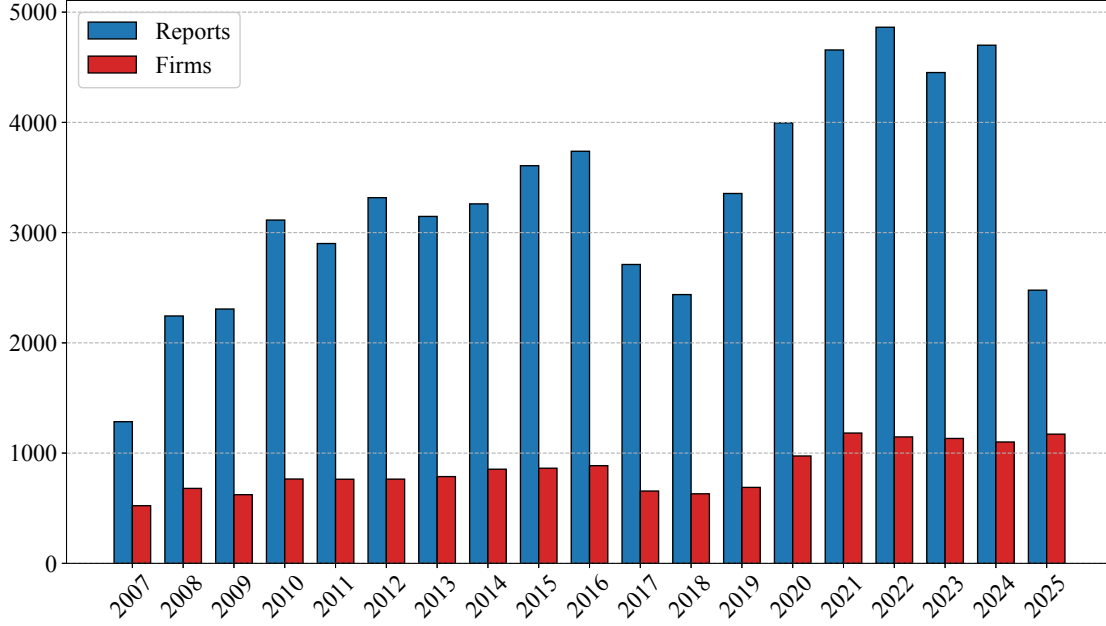


Figure 1 Histogram of Reports and Firms by Year.

Note: This figure shows the yearly count of Morgan Stanley Risk-Reward analyst reports and covered firms from January 2007 to May 2025.

whether analysts display a downside–upside asymmetry in the types of risks they emphasize across scenarios. To quantify the content of these narratives, we apply the GPT prompt described in Appendix Section A using the gpt-4o-mini model. Our main finding is that analysts place significantly greater emphasis on systematic downside risk in bear scenarios.

Figure 2 presents the report-level distributions of risk attention across scenarios. Panel A shows the distribution of attention to macro-level risks in bull, base, and bear cases. Panel B reports the corresponding distribution for firm-level risks, and Panel C does the same for industry-level risks. The figure reveals a clear pattern: relative to bull and base scenarios, bear scenarios assign substantially greater weight to macro-level risks and significantly less weight to firm-level risks. By contrast, the distributions of macro-level attention in bull and base scenarios are similar to each other, as are the distributions of firm-level attention across these two scenarios. Industry-level attention shows broadly similar distributions across all three scenarios.

Figure 3 complements this distribution evidence by showing the time-series evolution of

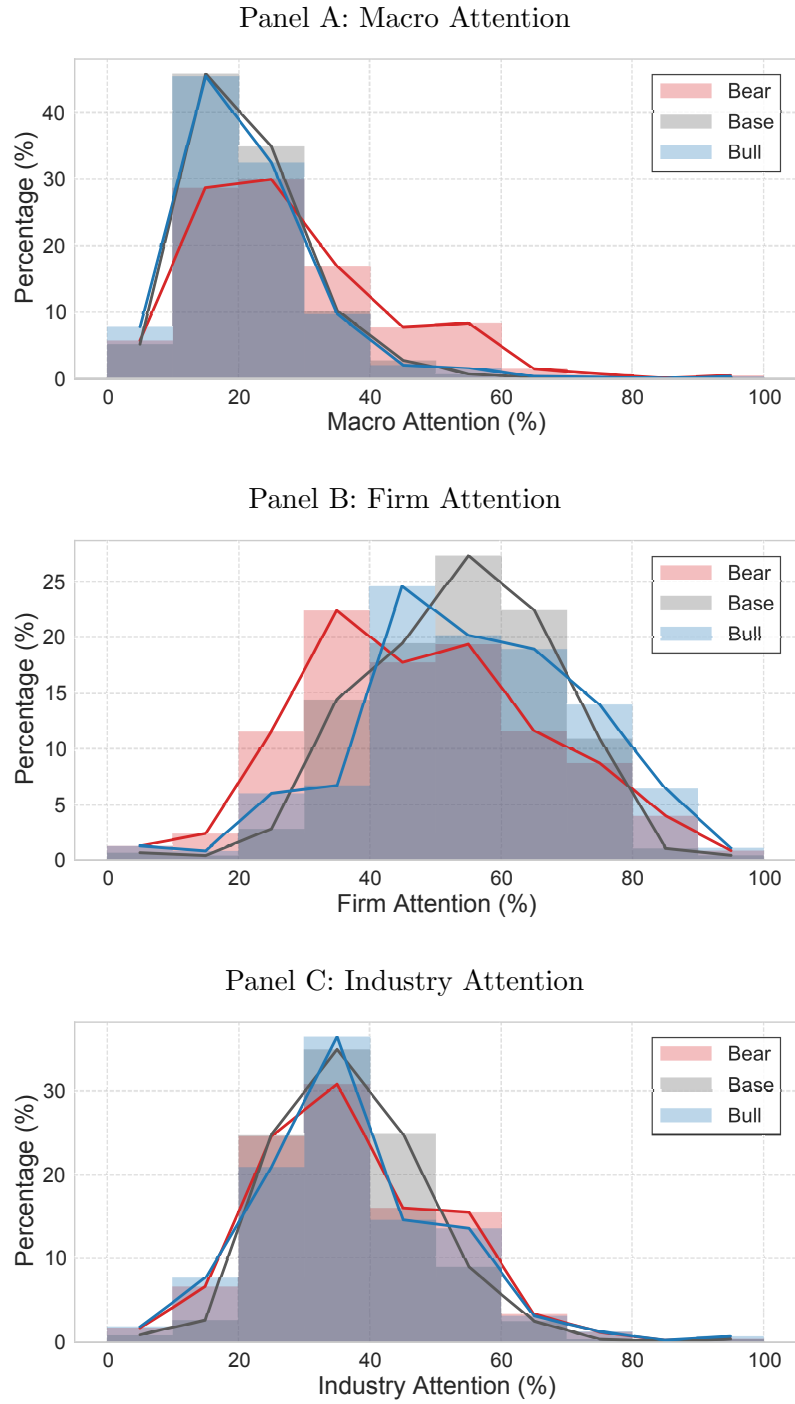


Figure 2 Attention Distribution to Macro, Firm, and Industry News.

Note: Report-level distribution of attention shares to firm-, industry-, and macro-level news across bull, base, and bear cases.

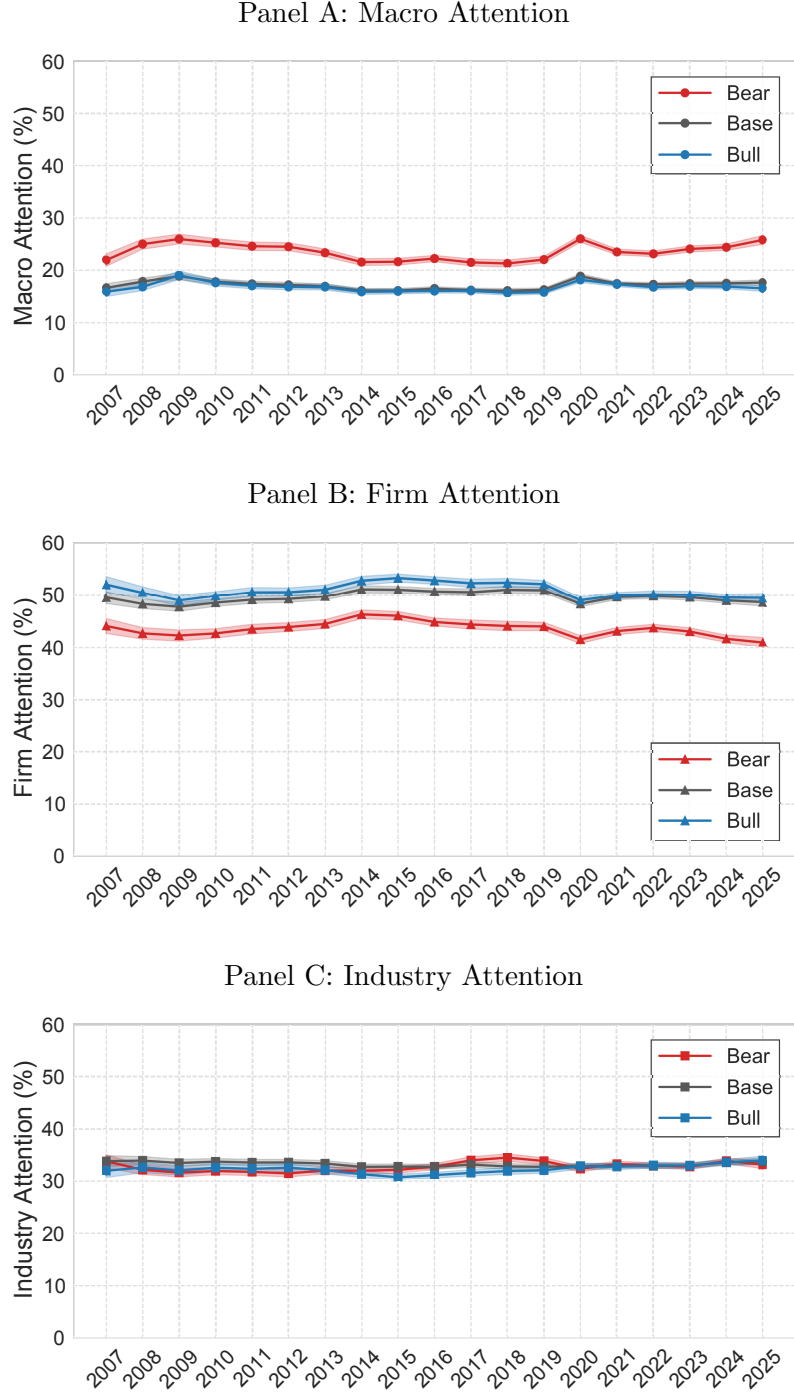


Figure 3 Time Series of Average Attention to Macro, Firm, and Industry News.

Note: Time series of report-level annual average attention shares to macro-, firm-, industry-level news. The shaded areas are 99% confidence interval of the average.

average attention to macro-, firm-, and industry-level risks from 2007 to 2025. Panel A plots macro-level attention, Panel B plots firm-level attention, and Panel C plots industry-level attention. The time-series evidence reinforces the distributional results: throughout the sample period, analysts consistently devote more attention to macro-level risks in bear scenarios than in bull or base scenarios, while placing less attention on firm-level risks in bear scenarios. In contrast, attention patterns in bull and base scenarios remain broadly similar over time for both macro- and firm-level risks. Industry-level attention again appears relatively stable and similar across bull, base, and bear cases. For completeness, Figure [A.1](#) reports the distribution at each point in time.

2.3 Realized CAPM R^2

To assess whether analysts are correct to place greater weight on macro risks in bear scenarios than in base or bull scenarios, we construct an *ex post* realized benchmark based on the realized CAPM R^2 in each case. The intuition is straightforward: if macro risks truly matter more in firms' downside states, then a firm's realized return variation should be more strongly explained by market-wide movements when the report ultimately realizes a bear outcome than when it realizes a base or bull outcome. Accordingly, we test whether the realized CAPM R^2 is significantly higher for realized-bear reports than for realized-base or realized-bull reports. A higher realized R^2 in bear outcomes would support analysts' emphasis on macro risks in downside scenarios; otherwise, it would suggest that analysts overweight macro risks in bear-case narratives.

Our empirical procedure has three steps. First, for each report, we compute the realized *ex post* CAPM R^2 by regressing the firm's daily excess returns on daily market excess returns over the 252 trading days following the report release. This R^2 measures the extent to which the firm's subsequent stock return variation is explained by systematic market movements.

Second, we classify each report as realized-bear, realized-base, or realized-bull according to the relationship between the analyst's target prices and the stock's realized price one year after the report release. A report is classified as realized-bear if the realized price is less than

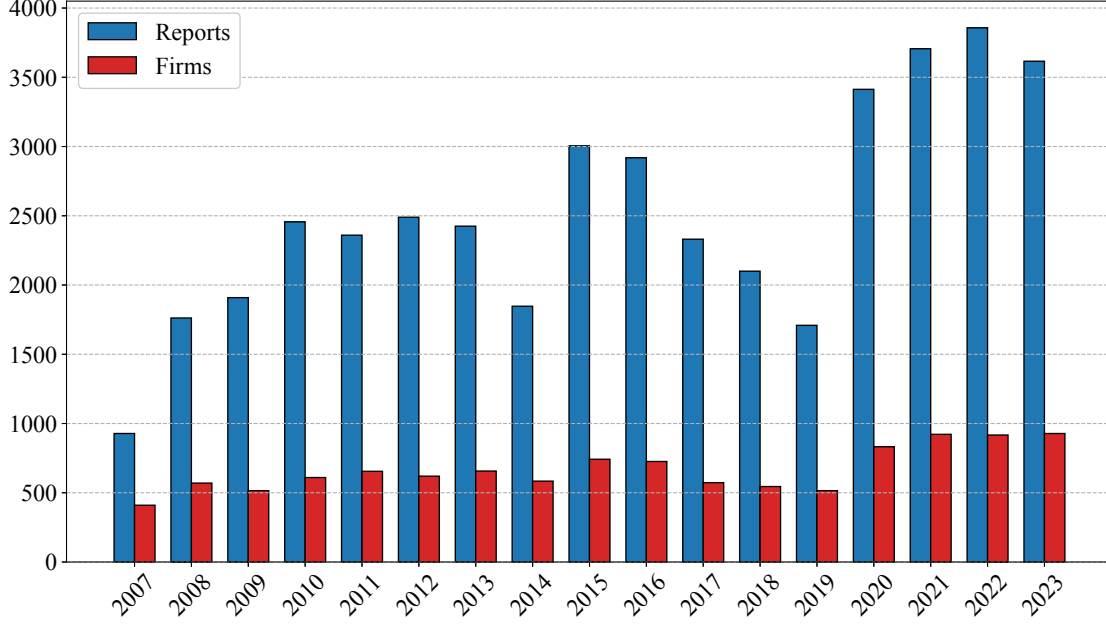
or equal to the bear-case target price. It is classified as realized-bull if the realized price is greater than or equal to the bull-case target price. If the realized price lies strictly between the bear and bull targets, the report is classified as realized-base.

Third, we aggregate these report-level realized R^2 measures to the firm level. For each firm, the realized-bear R^2 is the average realized CAPM R^2 across all of its realized-bear reports. Likewise, the realized-base and realized-bull R^2 are defined as the average realized CAPM R^2 across the firm’s realized-base and realized-bull reports, respectively. Constructing the measures at the firm level allows us to compare whether the realized R^2 is higher in bear outcomes than in base or bull outcomes for the same firm, thereby reducing concerns that cross-firm differences drive the results.

To make these comparisons within a common market environment, we further construct firm-level realized R^2 measures separately by market condition. Specifically, under a given market state, we restrict the sample to firms that experience all three realized report types—bear, base, and bull—within that same market condition. For example, when analyzing periods in which the market excess return over the 252 trading days after the report release lies between -5% and 5% , we include only firms that have realized-bear, realized-base, and realized-bull reports within that market-return range. This restriction ensures that comparisons across realized cases are not mechanically driven by differences in overall market environments.

We obtain stock return data from the CRSP database. After matching Morgan Stanley reports to 252-trading-day forward returns following the report release date, the sample contains 45,348 reports in total. Figure 2.3 presents the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms in the matched sample.

To mitigate distortions from stock splits and reverse splits, we construct an alternative measure of the one-year-ahead realized price by compounding daily returns excluding



Note: This figure shows the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms after matching the reports with one-year-ahead firm returns.

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Figure 4 Yearly Distribution of Reports and Firms after Matching with Returns.

Note: This figure shows the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms after matching the reports with one-year-ahead firm returns.

dividends, denoted by $r_{t,x}$, over the 252 trading days after the report release:

$$\text{constructed realized price} = \text{current price} \times \prod_{t=1}^{252} (1 + r_{t,x}).$$

If the realized price in one year differs from the constructed realized price by more than 10% in either direction, we replace the observed realized price with the constructed realized price. This adjustment helps correct cases in which stock splits or reverse splits mechanically distort the reported price level.

Table 1 Summary Statistics

Note: This table reports summary statistics for the Morgan Stanley Risk-Reward report sample from 2007 to 2025. Panel A reports report-level attention shares (percentages) by scenario. Panel B reports firm characteristics at the report date. All attention shares are extracted by GPT using the prompt in Appendix A.

Panel A: Narrative Attention								
Variable	N	Mean	Median	Std	Min	P25	P75	Max
Bull: Macro	45348	16.70	10.00	11.11	0.00	10.00	20.00	100.00
Bull: Industry	45348	32.28	30.00	14.32	0.00	20.00	40.00	100.00
Bull: Firm	45348	51.01	50.00	17.31	0.00	40.00	60.00	100.00
Base: Macro	45348	17.02	15.00	9.46	0.00	10.00	20.00	100.00
Base: Industry	45348	33.13	30.00	11.47	0.00	25.00	40.00	100.00
Base: Firm	45348	49.85	50.00	14.41	0.00	40.00	60.00	100.00
Bear: Macro	45348	23.35	20.00	15.11	0.00	10.00	30.00	100.00
Bear: Industry	45348	32.62	30.00	14.04	0.00	20.00	40.00	100.00
Bear: Firm	45348	44.03	40.00	18.32	0.00	30.00	60.00	100.00
Panel B: Returns and Forecast Variables								
Return	45348	0.12	0.07	0.74	-3.29	-0.17	0.31	74.14
Subjective Return	45348	0.18	0.10	1.43	-4.67	-0.01	0.22	97.63
Forecast Error	45348	-0.06	-0.02	1.59	-96.60	-0.30	0.26	74.03
$\hat{\beta}_{\text{CAPM}}$	42626	1.13	1.08	0.44	-1.04	0.83	1.36	5.29
R^2_{CAPM}	42626	0.32	0.30	0.18	0.00	0.18	0.45	0.86

Table 1 reports summary statistics of core narrative and firm-level data. In Panel A highlights the narrative asymmetry across scenarios. Bear-case macro attention averages 23.4%, compared with 17.0% and 16.7% in the base and bull cases, respectively—a bear–base gap of approximately 6.3 percentage points. The mirror image appears in firm-level attention: bear-case firm attention averages 44.0%, roughly 6 percentage points below the base-case mean of 49.9%, while industry attention is nearly identical across all three scenarios. Panel B shows that analyst base-case targets are slightly optimistic on average (mean forecast error of -0.06), and that the coverage universe skews toward high-beta firms (mean $\hat{\beta}_{\text{CAPM}} = 1.13$), consistent with Morgan Stanley’s large-capitalization focus.

Figure 5 illustrates how realized bear, base, and bull reports are distributed across

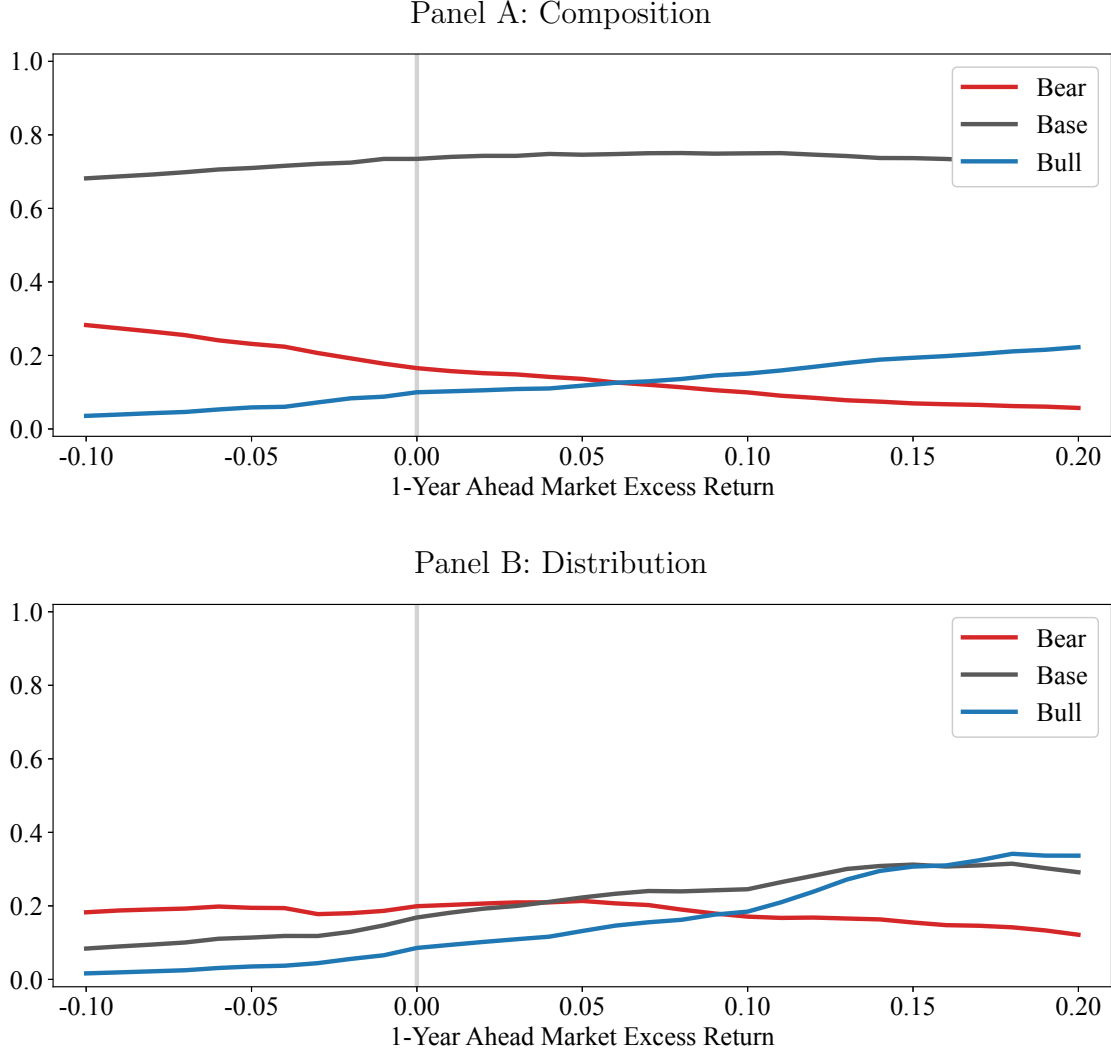


Figure 5 Composition and Distribution of Realized Bear, Base, and Bull Reports Across Market States.

Note: This figure plots the composition and distribution of realized bear, base, and bull reports across market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. Panel A shows the composition of bear, base, and bull reports within each return state:

$$\frac{\text{Number of realized bear/base/bull reports in a return state}}{\text{Total number of reports in that market excess return}}.$$

Panel B shows the distribution for realized bear, base, and bull reports:

$$\frac{\text{Number of realized bear/base/bull reports in a market excess return state}}{\text{Total number of realized bear/base/bull reports}}.$$

different market states. To characterize market conditions, we group reports according to the forward 252-day market excess return following the analyst report release. For each point on the horizontal axis, we consider reports whose forward market excess returns fall within a centered 10% window around that point.

The two panels present the same data from complementary perspectives. Panel A focuses on the composition of realized outcomes within each market state. For a given market-return state, it reports the shares of realized bear, base, and bull reports among all reports that fall into that state. This perspective shows how the relative likelihood of each realized outcome changes as aggregate market conditions become weaker or stronger. The figure shows a clear pattern: as future market excess returns rise, the share of realized bear reports declines, while the share of realized bull reports increases. In contrast, the share of realized base reports remains comparatively stable across market states. This pattern is intuitive. When the overall market performs poorly, more stocks are likely to end up near analysts' downside scenarios, making realized bear outcomes more common. When the market performs strongly, more stocks are likely to reach their upside scenarios, increasing the frequency of realized bull outcomes.

Panel B instead focuses on how each realized outcome is distributed across market states. For realized bear, base, and bull reports separately, it plots the fraction of that report type that falls into each market-return state. This perspective highlights the market environments in which each realized outcome is most likely to occur. Realized bear reports are spread more broadly across market states, resulting in a relatively flat profile. By contrast, realized base reports, and especially realized bull reports, become more concentrated in stronger market states. This pattern suggests that the distribution of forward market excess returns is tilted toward positive values, so realized base and bull outcomes are more likely to occur when subsequent market conditions are favorable.

Taken together, the two panels provide a more complete picture of the relation between realized report outcomes and subsequent market conditions. The figure shows that realized outcomes are systematically linked to the broader market environment: realized bear

outcomes are more prevalent when subsequent market conditions are weak, while realized bull outcomes become both more common and more concentrated when subsequent market conditions are strong. This implies that, when comparing realized bear, base, and bull benchmarks, it is important to condition on market state; otherwise, observed differences may conflate firm-level realization with variation in aggregate market performance.

2.4 Macro Bear Bias

Add definition

In this section, we examine whether analysts are justified in placing substantially greater attention on macro risks in bear cases than in base or bull cases. We provide evidence that analysts overweight systematic risk in downside scenarios.

To ensure an apples-to-apples comparison with realized R^2 , we construct firm-level ex ante macro attention for each case (bear, base, and bull) as the average macro attention across all reports of the firm that fall within the same market state. Thus, the comparison with the ex post realized R^2 is based on the exact same subset of reports.

Figure 6 plots realized R^2 and macro attention conditional on market state. For each point x on the horizontal axis, we consider reports whose forward 252-day market excess returns fall within a centered 10% window, $[x - 5\%, x + 5\%]$, around that point. Panel A shows the average CAPM R^2 over the 252 trading days after report release for realized bear (red), realized base (yellow), and realized bull (green) cases. The figure shows that, conditional on the same market state, the three realized R^2 series are very similar. At the same time, all three decline as subsequent market excess returns become weaker. This pattern indicates that exposure to systematic risk is primarily driven by the realized market environment, rather than by whether the report is ultimately classified as realized bear, base, or bull. Panel B shows the average ex ante macro attention assigned by analysts to the bear, base, and bull cases.

Two findings stand out. First, the three attention series are relatively flat across market

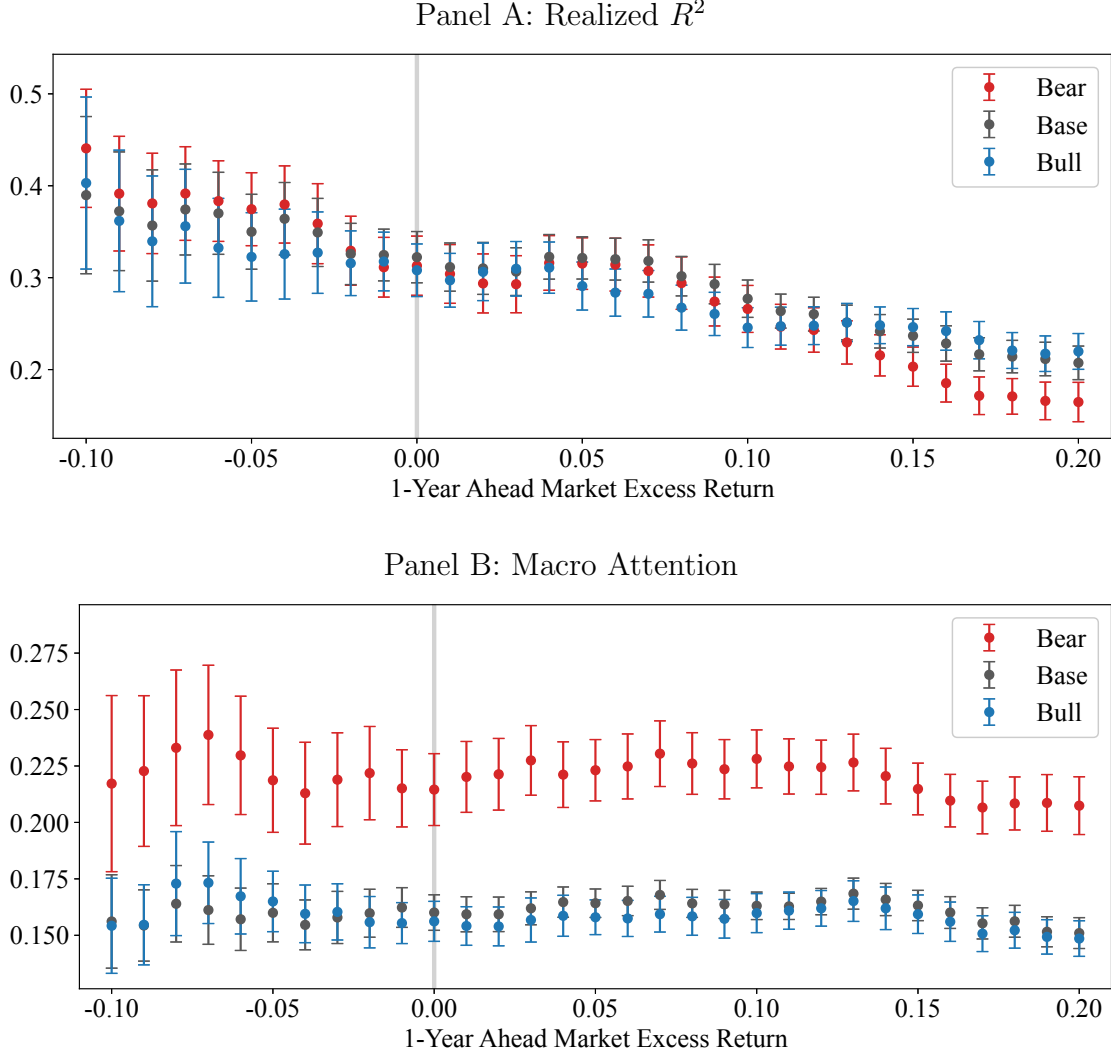


Figure 6 Realized R^2 and Macro Attention Conditional on Market States.

Note: This figure shows the realized R^2 and macro attention conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention within each realized scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

states compared with their realized R^2 benchmarks. This suggests that analysts do not sufficiently adjust their attribution of macro risk to the realized market environment.

Second, macro attention in the bear case is consistently and substantially higher than in the base or bull case. Figure 7 makes this pattern especially clear by plotting the bear–base

gap in macro attention together with the corresponding bear–base gap in realized R^2 across market states. The realized R^2 gap is close to zero when the subsequent annual market excess return is below 12%, and turns negative when the market excess return is above 12%. In contrast, the bear–base gap in macro attention remains significantly positive throughout. In other words, analysts consistently describe the bear case as much more macro-driven than the base case, even though the realized difference in systematic risk is very small or even negative, depending on the market state. This pattern provides direct evidence that analysts overweight systematic risk in downside narratives.

These patterns point to two possible explanations. The first is that analysts expect realized systematic risk to be roughly constant across market states, while at the same time systematically assigning extra weight to macro risk in bear cases. Under this explanation, analysts do condition on downside scenarios, but they fail to recognize that the importance of systematic risk varies with the market environment.

The second possibility is that analysts understand that realized R^2 declines with market returns, but still attach a persistently high probability to a severely negative future market outcome regardless of the prevailing market state. Under this interpretation, the elevated macro attention in bear cases would reflect a belief that downside scenarios are closely tied to large aggregate market declines.

The evidence is more consistent with the first explanation. If the second explanation were correct, two implications should follow. First, macro attention in the base case should move with market state and roughly track the realized base-case R^2 , which declines as subsequent market conditions improve. Second, macro attention in the bull case should be materially lower than in the base case, since that explanation implies analysts understand that realized R^2 falls with market returns and should therefore assign less macro risk to more favorable scenarios. The data support neither implication. Macro attention in the base case remains largely flat across market states, rather than tracking the realized base-case R^2 . In addition, conditional on the same market state, the realized R^2 series for the base and bull cases are similar, which is difficult to reconcile with the assumption that analysts meaningfully

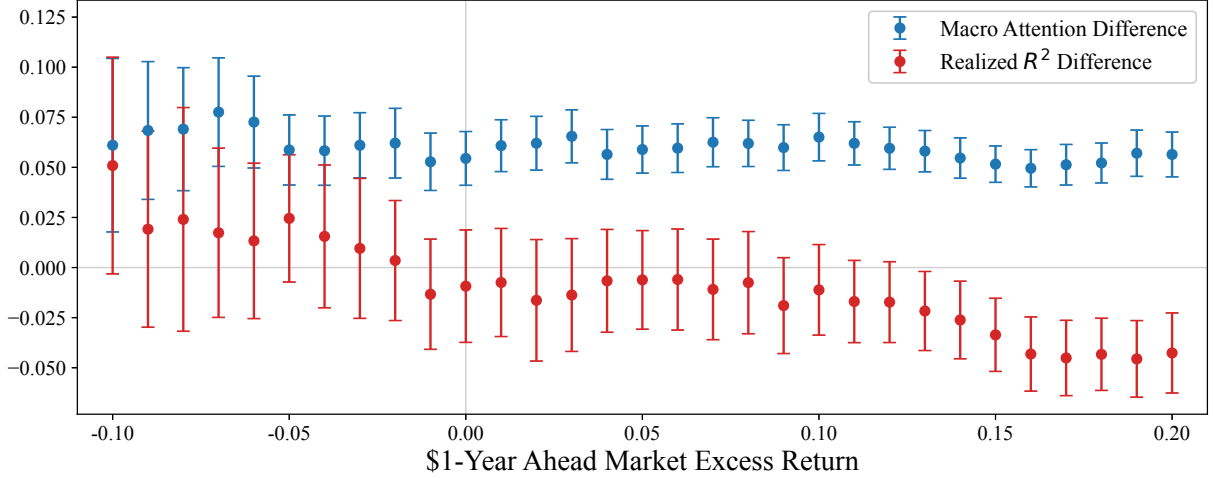


Figure 7 Bear – Base Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in macro attention and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bear and base cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

incorporate the relation between realized R^2 and market returns. Taken together, these findings suggest that analysts are not merely mapping different scenarios into different expected market states; rather, they appear to systematically overweight macro risk in downside narratives.

Explanation 1 more details: Analysts recall full sample or salient bear realization, instead of bear realizations’ underlying specific market conditions.

2.5 Conceptual Framework

Analysts construct bear-case scenarios using an implicit belief about the source of downside risk. We formalize this belief as a subjective probability that the bear outcome is driven by macroeconomic rather than firm-specific forces. For firm i at report date t , let $\text{MacroAttention}_{i,t}^{\text{Bear}}$ denote the percentage of bear-case narrative text attributed to macroeconomic factors (defined formally in Section 2.1), and define the analyst’s implicit macro

attribution probability as $\hat{p}_{\text{Bear},i,t}^{\text{macro}} \propto \text{MacroAttention}_{i,t}^{\text{Bear}}$. A well-calibrated analyst would set this probability as a function of the market state. When markets fall, realized bear-case outcomes are predominantly systematic—firms declining together with the market—so the true probability $p_{\text{Bear}}^{\text{true}}$ should be high. When markets rise, bear-case realizations are predominantly idiosyncratic—firms suffering company-specific collapses against an otherwise buoyant backdrop—so $p_{\text{Bear}}^{\text{true}}$ should be low.

A misattribution bias may arise when $\hat{p}_{\text{Bear},i,t}^{\text{macro}}$ fails to track this state-dependence. The key prediction is that a miscalibrated analyst applies the macro-heavy bear template unconditionally: $\hat{p}_{\text{Bear},i,t}^{\text{macro}}$ remains elevated relative to base-case macro attention in both rising and falling markets, even as realized systematic risk varies substantially with market conditions. The result is a subjective conditional model applied unconditionally. We test this prediction in Section 2.4.

The bias is a miscalibration of *source*, not magnitude. The bear-case target price may well reflect a reasonable assessment of the magnitude of downside risk; the error lies in attributing that downside to macroeconomic forces regardless of market conditions. In down markets the attribution is approximately correct—bear-case realizations are disproportionately driven by systematic forces. In up markets it is not: firms that realize their bear case do so through idiosyncratic disasters (earnings collapses, management failure, competitive disruption), yet the bear narrative invokes macro factors with equal frequency. This framing connects directly to [Ang and Chen \(2002\)](#), who document that U.S. stocks comove substantially more with the market during downturns: this misattribution bias is the narrative over-generalization of this genuine empirical regularity, treating a regime-dependent feature of market environments as an unconditional property of bear-case construction.

The asset-pricing consequences of this misattribution operate through investors. [Joos, Piotroski, and Srinivasan \(2016\)](#) document that Morgan Stanley base-case price targets are reflected in market prices; we propose that the *source attribution* in the bear narrative, not only the target price, shapes investor beliefs about the nature of downside risk. A macro-framed bear narrative leads investors to perceive the stock’s downside as contingent

on a broad macro shock—making that risk appear priced and hedgeable—while leaving idiosyncratic tail risk underweighted regardless of the macro environment. When this framing is over-applied, idiosyncratic tail risk is systematically underpriced, generating the forecast errors and portfolio abnormal returns we examine in Section 3.

3 Asset Pricing Implications

The core interpretation of the finding is: Analysts overestimate the importance of systematic (macro/market) risk relative to idiosyncratic risk in downside risk.

3.1 Analyst Forecast Error Prediction

We next examine whether downside overweighting of systematic risk predicts errors in analysts' base-case forecasts. Define the realized return and the analyst's base-case forecasted return as

$$R_{i,t+1}^{\text{realized}} \equiv \frac{P_{i,t+1}^{\text{realized}}}{P_t}, \quad R_{i,t+1}^{\text{base}} \equiv \frac{P_{i,t+1}^{\text{base}}}{P_t}.$$

We define the base-case forecast error as

$$\text{Forecast Error}_{i,t+1} = R_{i,t+1}^{\text{realized}} - R_{i,t+1}^{\text{base}}.$$

Under this definition, a positive forecast error indicates that the analyst's base-case forecast is too pessimistic relative to the subsequently realized return.

To formalize the role of downside overweighting, suppose realized returns satisfy

$$R_{i,t+1}^{\text{realized}} = \alpha_{i,t} + \beta_i M_{t+1} + \varepsilon_{i,t+1},$$

where $M_{t+1} = R_{t+1}^M - R_{t+1}^f$ denotes the future market excess return, β_i is firm i 's market exposure, and $\varepsilon_{i,t+1}$ captures firm-specific news. In the absence of bias, the analyst's base-

case forecast would be

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t,$$

where $\mu_t = E_t[M_{t+1}]$ is the analyst's expectation of the future market return.

To capture downside overweighting of systematic risk, define the macro attention gap between bear case and base case as

$$\text{MacroGap}_{i,t} = \text{MacroAttention}_{i,t}^{\text{bear}} - \text{MacroAttention}_{i,t}^{\text{base}},$$

and suppose analysts distort the base-case forecast downward in proportion to this bear–base macro-attention gap:

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t - \theta \beta_i \text{MacroGap}_{i,t}, \quad \theta > 0.$$

Under this specification,

$$\text{Forecast Error}_{i,t} = \beta_i (M_{t+1} - \mu_t) + \theta \beta_i \text{MacroGap}_{i,t} + \varepsilon_{i,t+1}.$$

Proposition. If analysts overweight systematic downside risk when forming their narratives, then the bear–base macro-attention gap positively predicts the base-case forecast error.

Intuition. A larger bear–base macro-attention gap reflects greater emphasis on macro downside risk in analysts' narratives. If analysts overweight systematic downside risk when forming these narratives, they overestimate the importance of macro downside risk and, as a result, set a lower base-case price forecast. This downward distortion makes the forecasted return too low relative to the subsequently realized return, thereby generating a larger forecast error.

Hypothesis. Firms with a larger bear–base macro-attention gap have larger subsequent base-case forecast errors:

$$\frac{\partial \text{Forecast Error}_{i,t+1}}{\partial \text{MacroGap}_{i,t}} > 0.$$

This prediction leads to the following empirical specification to predict analysts’ forecast error:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \text{MacroGap}_{i,t} + b_2 M_{i,t} + \Gamma' X_{i,t} + \gamma_i + u_{i,t+1}, \quad (1)$$

where $X_{i,t}$ includes firm-level controls, γ_i denotes firm fixed effects, and $M_{i,t}$ is the past 252-day market excess return prior to the report release. In our baseline specification, $X_{i,t}$ contains only the stock’s own past 252-day return before the report release. We include $M_{i,t}$ to control for the market state prevailing when the analyst issues the report. Hypothesis 1 predicts $b_1 > 0$.

For the empirical analysis, we go beyond the bear–base macro-attention gap and also examine the bull–base macro-attention gap, as well as the corresponding attention gap to firm-specific risks as a placebo test. This design helps distinguish our mechanism from more general differences in narrative intensity across scenarios. If forecast errors are specifically driven by downside overweighting of systematic risk, then the relevant predictor should be the bear–base gap in macro attention, rather than scenario differences in upside narratives or in firm-level risk attention.

The regression results are reported in Table 2. Across columns (1), (5), and (6), the coefficient on the bear–base macro-attention gap is positive and statistically significant, with t -statistics of 2.37, 2.07, and 3.25, respectively. These results indicate that firms for which analysts place relatively greater macro emphasis in the bear case than in the base case subsequently exhibit larger base-case forecast errors. Under our definition of forecast error, this means that analysts set base-case forecasts that are too low for these firms. The evidence

is therefore consistent with the hypothesis that downside overweighting of systematic risk distorts analysts' base-case expectations in a systematically pessimistic direction.

Table 2 Forecast Error Regressions

Note: This table reports empirical results for forecast error prediction regression (1). Columns (1)–(4) include each gap individually. Column (5) includes the two macro gaps jointly. Column (6) includes all four narrative gap variables simultaneously. Firm fixed effects are included in all specifications. Standard errors are clustered by firm and month. t-statistics are reported in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Attention Gaps						
Macro Bear – Base	0.0012** (2.37)				0.0011** (2.07)	0.0023*** (3.25)
Macro Bull – Base		0.0009 (1.03)			0.0005 (0.53)	0.0002 (0.23)
Firm Bear – Base			0.0006 (1.10)			0.0018** (2.28)
Firm Bull – Base				-0.0002 (-0.25)		-0.0004 (-0.57)
Control Variables						
Market Excess Return	-0.1467 (-1.56)	-0.1477 (-1.57)	-0.1474 (-1.57)	-0.1473 (-1.57)	-0.1469 (-1.56)	-0.1470 (-1.57)
Firm Excess Return	-2.3962 (-0.15)	-2.3370 (-0.14)	-2.1720 (-0.13)	-2.3007 (-0.14)	-2.4154 (-0.15)	-2.1880 (-0.13)
Observations	43,313	43,313	43,313	43,313	43,313	43,313
R^2	0.0004	0.0003	0.0003	0.0003	0.0004	0.0006

3.2 Nonlinear Analyst Forecast Error Prediction

Table 2 shows that the linear specifications have very limited explanatory power: the in-sample R^2 ranges only from 0.04% to 0.06%, despite the fact that the macro attention gap between the bear and base cases significantly predicts subsequent forecast error. This pattern suggests that the predictive content in analyst narratives is not well captured by a simple linear specification. While the bear–base macro attention gap is informative, its

effect appears too limited and state-dependent to generate substantial fit in an additive linear model.

This interpretation is consistent with the mechanism developed in Section 2.4. There, we show that analysts tend to overweight systematic risk in downside narratives without sufficiently conditioning on the prevailing market environment. As a result, the mapping from attention gaps to forecast errors should vary across market states. The same bear–base macro attention gap may therefore imply different forecast errors depending on whether aggregate conditions are weak or strong. This state dependence implies that forecast error is likely governed by nonlinear interactions between narrative-based attention measures and market conditions, rather than by their separate linear effects.

To accommodate this nonlinearity, we follow Kelly, Malamud, and Zhou (2024) and employ the random Fourier features (RFF) approach of Rahimi and Recht (2007, 2008). Our raw predictors consist of the macro bear–base gap, macro bull–base gap, firm bear–base gap, firm bull–base gap, and the previous 252-day market excess return prior to report release. We exclude lagged firm-level excess returns so that firm-specific information enters only through the attention measures extracted from analyst reports.

Let G_t denote the 5×1 predictor vector for a given firm-date observation. The RFF procedure maps G_t into nonlinear features,

$$S_{i,t} = [\sin(\gamma\omega_i'G_t), \cos(\gamma\omega_i'G_t)]', \quad \omega_i \sim i.i.d.N(0, I), \quad (2)$$

where each ω_i defines a random projection of the original predictors, which is then transformed through sine and cosine functions. This transformation generates a rich set of nonlinear combinations of attention gaps and market states. As emphasized by Kelly, Malamud, and Zhou (2024), RFF can be viewed as a wide two-layer neural network with random first-layer weights and estimated second-layer coefficients. Its main advantage in our setting is that it introduces flexible nonlinearity while remaining computationally tractable and straightforward to evaluate out of sample.

Following Kelly, Malamud, and Zhou (2024), we use a rolling training window of $T = 12$ and consider models with up to $P = 12,000$ random Fourier features. We set the bandwidth parameter to $\gamma = 2$. For each firm, we generate 12,000 RFFs, estimate recursive out-of-sample forecasts over a grid of model sizes $P \in \{2, \dots, 12,000\}$ and ridge penalties $\log_{10}(z) \in \{-3, \dots, 3\}$, and compute the resulting out-of-sample R^2 . Each firm is forecasted separately using only its own attention-gap variables and the aggregate market state, and the overall out-of-sample R^2 is then computed from the pooled set of firm-level predictions.

Figure A.3 reports the out-of-sample R^2 across model complexity levels. Consistent with Kelly, Malamud, and Zhou (2024), predictive performance initially declines as complexity increases, reaches its trough around $c = 1$, and then improves as complexity rises further. Figure 7 provides a zoomed-in view of the out-of-sample R^2 over the range from -0.05 to 0.05 . The out-of-sample R^2 peaks at approximately 4% when $c = 1000$. Relative to the near-zero in-sample R^2 of the linear model, this improvement is substantial. Overall, the results point to a strongly nonlinear predictive relation: analysts' attention gaps contain economically meaningful information, but that information is only effectively captured by a flexible nonlinear specification.

3.3 Portfolio Construction

We evaluate the portfolio implications of analyst forecasts using two sorting signals. The first signal is the predicted return implied by the analysts' base case forecast. The second signal is a bias-adjusted predicted return, constructed as the sum of the analyst-predicted return and the out-of-sample RFF forecast error estimated from the RFF regression.³ This adjusted measure is intended to capture both the analyst's baseline expectation and the predictable component of forecast bias.

For each sorting signal, we form long-short portfolios using an annual rebalancing strategy implemented separately at the end of each calendar month. Specifically, for each month of the year, we construct a portfolio that is rebalanced annually in that month, which yields

³We choose shrinkage z as 10 and complexity level $c = 1000$ in portfolio sorting.

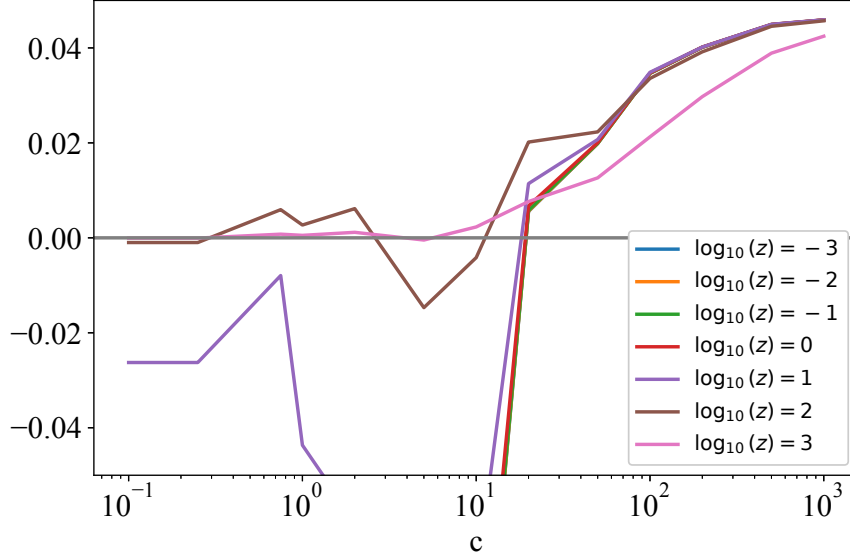


Figure 8 Out-of-sample Market Timing Performance (T=12).

Note: This figure shows the out-of-sample R^2 that use RFFs to predict one-year ahead forecast error. The training window is $T = 12$ reports and RFF count P (or cT) ranges from 3 to 12,000 with $\gamma = 2$. The horizontal axis is the complexity level $c = P/T$.

12 distinct annually rebalanced strategies. We then compute the average return for each of the 12 long-short portfolios and report their average alpha.

To examine how portfolio performance varies with the strength of the sorting signal, we consider several alternative portfolio cutoffs. In particular, we implement tercile-style sorts by going long the top 30% of stocks and short the bottom 30%, quintile-style sorts by going long the top 20% and short the bottom 20%, and decile-style sorts by going long the top 10% and short the bottom 10%. For each specification, we construct both equal-weighted and value-weighted versions of the long-short portfolio.

Table 3 reports the average returns and CAPM alphas of the portfolio strategies. Panel A sorts stocks on the analyst base-case forecast return and thus provides the baseline results. The corresponding long-short portfolios generate weak performance: average returns are generally insignificant or only marginally significant at the 10% level. This finding suggests that the analysts' base-case forecast alone has limited power to identify subsequent abnormal returns.

Panel B instead sorts stocks on the bias-adjusted signal, defined as the base forecast

return plus the out-of-sample predicted forecast error from the RFF model. Portfolio performance improves sharply under this specification. Both average returns and CAPM alphas are positive and statistically significant at the 1% level. In the decile sort, the equal-weighted long-short portfolio delivers a monthly CAPM alpha of 1.65% with a t -statistic of 5.60, while the value-weighted portfolio delivers a monthly CAPM alpha of 1.53% with a t -statistic of 5.15. Since the baseline sort in Panel A generates little abnormal performance, the improvement in Panel B indicates that the predictive gains are driven primarily by the out-of-sample forecast-error component.

This result suggests that predictable analyst forecast bias is economically important. Adjusting for this bias substantially improves the cross-sectional ranking of stocks, implying that the predictable component of forecast errors contains information not fully incorporated into prices at the time of report issuance. The similarity between equal-weighted and value-weighted results further indicates that this predictive content is not confined to small firms, but extends to larger stocks as well.

As the sort moves from deciles to quintiles and terciles, average returns decline in magnitude. However, returns remain positive and significant at the 1% level. This pattern indicates that the profitability of the bias-adjusted signal is not driven solely by a small set of extreme stocks, but instead reflects a broad and robust source of cross-sectional predictability.

Figure 9 provides further support by plotting CAPM alphas across the 10 deciles for both equal-weighted (blue) and value-weighted (red) portfolios. In both cases, CAPM alpha declines monotonically across deciles. The top three deciles exhibit significantly positive alphas, whereas the bottom three deciles exhibit significantly negative alphas. This monotonic pattern indicates that the bias-adjusted signal does more than separate the extreme portfolios: it delivers a systematic ranking of stocks throughout the distribution by subsequent abnormal returns. More broadly, this result suggests that the nonlinear interaction between macro-attention gaps and market states contains meaningful predictive information about subsequent analyst forecast errors.

Table 3 Long-Short Portfolio Performance

Note: This table reports the performance of long-short portfolios formed on two sorting signals derived from analyst forecasts. Panel A sorts stocks on the predicted return implied by the analyst’s base-case forecast. Panel B sorts stocks on a bias-adjusted predicted return, defined as the analyst-predicted return plus the out-of-sample predicted forecast error from the RFF model. For each signal, portfolios are formed separately at the end of each calendar month and rebalanced annually, yielding 12 distinct annually rebalanced strategies. Reported returns and alphas are averages across the 12 strategies. Long-short portfolios buy stocks in the top 10%, 20%, or 30% of the sorting distribution and short stocks in the bottom 10%, 20%, or 30%, corresponding to decile-, quintile-, and tercile-style sorts. EW and VW denote equal- and value-weighted portfolios. Avg Return is the average monthly long-short return, and CAPM Alpha is the corresponding monthly CAPM alpha. Adjacent columns report t -statistics. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Long-Short	Weight	Avg Return	t -statistic	CAPM Alpha	CAPM t -statistic
Panel A: Base Forecast Return					
Top 10% - Bottom 10%	EW	0.0021	0.76	-0.0009	-0.39
Top 10% - Bottom 10%	VW	0.0028	0.96	0.0007	0.29
Top 20% - Bottom 20%	EW	-0.0061*	-1.75	-0.0097**	-2.41
Top 20% - Bottom 20%	VW	-0.0050	-1.37	-0.0082**	-1.97
Top 30% - Bottom 30%	EW	-0.0049*	-1.80	-0.0074**	-2.52
Top 30% - Bottom 30%	VW	-0.0036	-1.21	-0.0060*	-1.86
Panel B: Base Forecast Return + OOS Predicted Forecast Error					
Top 10% - Bottom 10%	EW	0.0177***	5.47	0.0165***	5.60
Top 10% - Bottom 10%	VW	0.0178***	5.19	0.0152***	5.15
Top 20% - Bottom 20%	EW	0.0150***	3.79	0.0173***	3.54
Top 20% - Bottom 20%	VW	0.0142***	3.46	0.0164***	3.28
Top 30% - Bottom 30%	EW	0.0133***	3.29	0.0157***	3.06
Top 30% - Bottom 30%	VW	0.0126***	3.07	0.0147***	2.82

4 Mechanisms

The preceding sections establish a systematic bias in analyst scenario construction: bear-case narratives assign substantially greater weight to macroeconomic factors than base-case narratives, and this narrative gap remains stable across all market environments even as the realized CAPM R^2 gap varies sharply with the market state. In this section, we investigate the underlying sources of this miscalibration. We first rule out two naive mechanisms and then propose two complementary behavioral channels—crisis anchoring via the availability

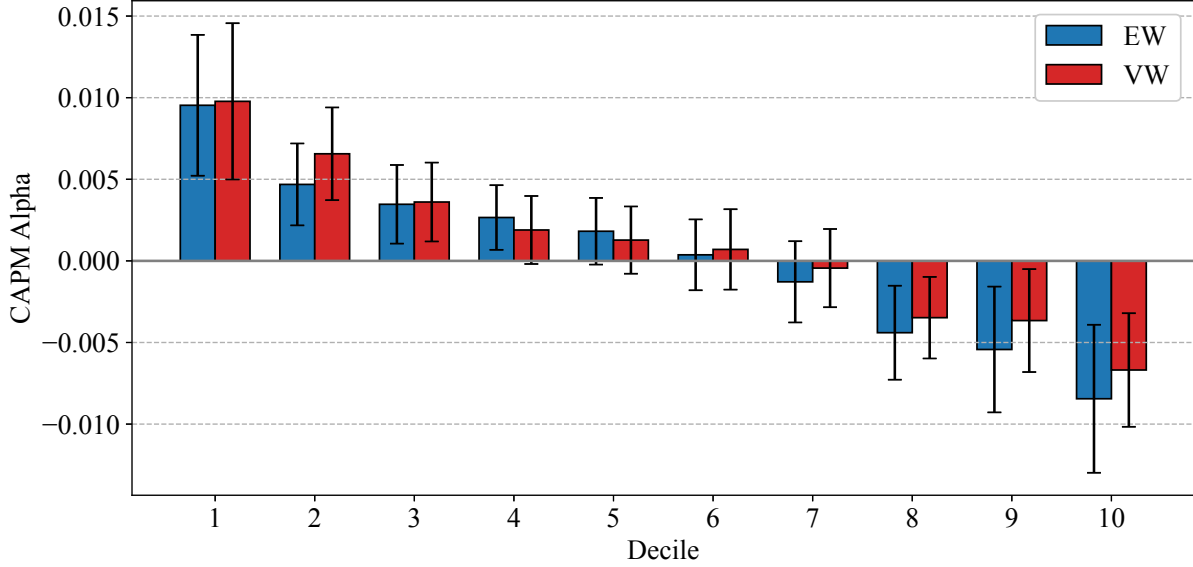


Figure 9 Portfolio CAPM Alpha in 10 Deciles.

Note: This figure reports monthly CAPM alphas for the 10 decile portfolios formed on the bias-adjusted predicted return, defined as the analyst-predicted return plus the out-of-sample predicted forecast error from the RFF model. Decile 1 contains stocks with the highest values of the sorting signal, and Decile 10 contains stocks with the lowest. Portfolios are formed using the same annual rebalancing procedure implemented separately at the end of each calendar month, and reported alphas are averages across the resulting 12 annually rebalanced strategies. Blue bars denote equal-weighted (EW) portfolios and red bars denote value-weighted (VW) portfolios. The monotonic decline in alpha across deciles indicates that the bias-adjusted signal delivers a systematic cross-sectional ranking of subsequent abnormal returns.

heuristic and career concerns that generate a cover-your-asset (CYA) heuristic—each of which is consistent with the documented evidence and generates distinct cross-sectional predictions.

Ruling out naive mechanisms. Two naive mechanisms could generate a persistent narrative asymmetry. The first is a flat- R^2 prior: analysts may believe that systematic risk is genuinely higher in bear states—a belief partially justified by the higher downside correlations documented by [Ang and Chen \(2002\)](#)—and therefore assign unconditionally greater macro attention to bear narratives. This explanation is consistent with the unconditional asymmetry (H1) but not with its state-independence. If analysts correctly perceived that systematic risk in bear-state outcomes varies with market conditions, macro attention in bear

narratives should decline in up markets, when realized bear-case outcomes are predominantly idiosyncratic. The flat narrative gap in Figure 7 contradicts this prediction.

The second naive mechanism is a persistent bear-market prior: analysts may chronically assign high probability to a severe macro downturn regardless of current conditions. This explanation predicts that macro attention would be elevated uniformly across all scenarios. In particular, it implies that base-case macro attention should exceed realized base-case R^2 , and that bull-case macro attention should fall meaningfully below the base case. Neither pattern is present in Figure 6: base-case and bull-case macro attention are nearly identical and flat across market states. The elevated macro attention is specific to bear narratives and cannot be attributed to a generalized pessimistic prior.

Availability heuristic and anchoring. The bear-narrative template is anchored to the stock of vivid macro catastrophes in collective analyst memory. The 2008 Global Financial Crisis and the 2020 COVID crash are the most salient exemplars of severe equity downturns: both were swift, extreme, and overwhelmingly systematic. The availability heuristic (Tversky and Kahneman, 1973) implies that analysts rely on such vivid instances when constructing downside scenarios, making macro-driven collapses the default bear-case template. Idiosyncratic disasters are diffuse, firm-specific, and narratively incoherent across analysts, and therefore less available as a shared template.

The viral character of macro narratives reinforces this anchoring (Shiller, 2019): recession narratives spread across the investment community and persist long after fundamentals normalize. Consistent with this account, Figure 3 shows that macro bear attention rises significantly around the COVID episode and remains elevated in the subsequent recovery, while the realized R^2 gap narrows considerably once markets stabilize (Figure 7). This divergence—slow decay of the narrative gap relative to faster normalization of the empirical benchmark—is the temporal signature of crisis anchoring. Cross-sectionally, the mechanism predicts narrative herding: analysts covering firms in the same industry should exhibit

correlated macro emphasis in bear narratives, as they draw from the same pool of recent crisis templates.

Career concerns and the cover-your-asset heuristic. Career concerns provide a complementary explanation for why analysts maintain macro-heavy bear narratives even when a firm’s downside risk is predominantly idiosyncratic. Sell-side analysts face asymmetric reputational costs: idiosyncratic errors are more visible and more damaging than consensus-correlated errors (Hong and Kubik, 2003), creating incentives to anchor bear narratives to shared macro risks.

A macro-driven bear narrative offers three advantages in this reputational calculus. First, it is non-falsifiable in real time: a miss can be attributed to an unexpectedly resilient macro environment rather than to a firm-specific judgment error. Second, it is relationally neutral: macro claims do not implicate management, reducing friction in coverage relationships. Third, it is consensus-aligned: anchoring to widely discussed macro risks provides cover from ex-post criticism. A firm-specific bear narrative, by contrast, is verifiable and implicitly accusatory.

This cover-your-asset (CYA) heuristic generates testable cross-sectional predictions. Analysts at brokerages with active investment-banking relationships should exhibit larger narrative-realization gap for covered firms, as the reputational cost of antagonizing management is higher. Junior analysts, facing greater downside from differentiated misses, should exhibit stronger CYA behavior than experienced peers. Conversely, analysts known for bold or contrarian calls should exhibit smaller narrative-realization gap.

Finally, the availability and career-concerns channels are complementary. Availability anchoring explains why macro narratives are the template analysts *reach for*; career concerns explain why they are *reluctant to deviate* from it. Together, they predict the macro bear bias is strongest during and after crisis episodes—when macro narratives are most available—and among analysts with stronger career or reputation concerns.

5 Conclusion

We document a systematic pattern in how analysts narrate downside risk. In Morgan Stanley Risk-Reward reports spanning 2007 to 2025, bear-case narratives assign roughly six percentage points more attention to macroeconomic factors than base- or bull-case narratives, a gap that is stable across market regimes, crisis episodes, and expansions alike. We formalize this as a macro bear bias—the overgeneralization of the genuine downside comovement documented by [Ang and Chen \(2002\)](#) into a state-independent narrative template. The bias is a miscalibration of source, not magnitude: the bear–base macro-attention gap does not co-vary with realized CAPM R^2 conditional on market state, even as the realized R^2 gap reverses sign from positive in down markets to negative in up markets. The misattribution has measurable downstream consequences. The bear–base attention gap positively predicts subsequent analyst forecast errors, with t -statistics above 3 in the most saturated specification, and is absent for the bull–base gap, confirming that downside overweighting is the operative channel. Portfolios formed on a nonlinear out-of-sample forecast-error correction earn monthly CAPM alphas of up to 1.65% ($t = 5.60$), compared to near-zero or negative alphas from the unadjusted analyst signal.

Two mechanisms plausibly account for the pattern. The availability heuristic ([Tversky and Kahneman, 1973](#)) predicts that vivid macro catastrophes—the 2008 financial crisis, the 2020 pandemic shock—anchor analysts’ default bear-case template and that this anchor decays slowly, generating macro-heavy narratives well into the subsequent recovery. Career concerns provide a complementary incentive channel ([Holmström, 1999](#); [Scharfstein and Stein, 1990](#)): macro-attributed downside is harder to falsify in real time and less reputationally costly than firm-specific attribution, making macro framing a strategically safe default. Both channels predict that the narrative gap widens during and immediately after macro crises.

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INTERNET APPENDIX

A LLM Prompt

You are a financial analyst trained in equity valuation and empirical asset pricing. You will analyze the Bull, Base, and Bear case texts from a Morgan Stanley equity research report. Your task is to extract reasoning, classify it as firm-level, industry-level, or macro-level, identify the economic channel, and quantify attention shares across cases. Write in a concise, research-style tone used in equity research reports.

Return your full answer ****strictly in JSON format****, with one JSON object per report (suitable for .jsonl storage). Each JSON object must include keys: "filename", "bull_case", "base_case", "bear_case", and "cross_case_summary". Do not include explanations outside the JSON.

Tasks:

1. Price Targets: List the bull case, base case, and bear case price targets stated in the Risk Reward section.
2. Reason Extraction: Identify all distinct reasons for the bull, base, and bear cases. Requirements:
 - 2.1 Include any explicitly stated upside risks as part of the bull-case reasons.
 - 2.2 Include any explicitly stated downside risks as part of the bear-case reasons.
 - 2.3 Preserve the original wording as closely as possible (avoid paraphrasing).
 - 2.4 Number each reason sequentially (e.g., 1., 2., 3.).
 - 2.5 Label each reason as one of the following categories: Firm, Industry, Macro, or Mixed. If labeled Mixed, specify which component (Firm, Industry, or Macro) is dominant.
 - 2.6 Assign each reason to a single economic channel that best represents

its underlying mechanism, such as expected cash flow, discount rate, risk aversion or asset demand.

- 3.Attention Quantification: Quantify the analyst's relative attention across Firm, Industry, and Macro news as: (Firm %, Industry %, Macro %), ensuring $\text{Firm \%} + \text{Industry \%} + \text{Macro \%} \leq 100$.
- 4.Attention Evolution Summary: Summarize how the analyst's attention distribution across Firm, Industry, and Macro news (from Step 2.5) evolves across the bull, base, and bear cases.
- 5.Channel Evolution Summary: Summarize how the dominant economic channels (from Step 2.6) evolve across the bull, base, and bear cases.

Output format (strict JSON):

```
{
  "filename": "<report_name>",
  "bull_case": {
    "bull_price_target": "<numerical price target or range>",
    "reasons": [
      {
        "id": 1,
        "text": "<verbatim reason>",
        "category": "Firm | Industry | Macro | Mixed (dominant: Firm/
          Industry/Macro)",
        "economic_channel": "Expected Cash Flow | Discount Rate | Risk
          Aversion | Asset Demand"
      }
    ],
    "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
    "analytical_summary": "[one concise sentence summarizing key focus of
      bull case]"
  },
  "base_case": {
    "base_price_target": "<numerical price target or range>",
```



```

"reasons": [
  {
    "id": 1,
    "text": "<verbatim reason>",
    "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
    "economic_channel": "Expected Cash Flow | Discount Rate | Risk
      Aversion | Asset Demand"
  }
],
"attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
"analytical_summary": "[one concise sentence summarizing key focus of
  base case]"
},

"bear_case": {
  "bear_price_target": "<numerical price target or range>",
  "reasons": [
    {
      "id": 1,
      "text": "<verbatim reason>",
      "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
      "economic_channel": "Expected Cash Flow | Discount Rate | Risk
        Aversion | Asset Demand"
    }
  ],
  "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
  "analytical_summary": "[one concise sentence summarizing key focus of
    bear case]"
},

"cross_case_summary": {
  "attention_evolution": "[summary of how Firm/Industry/Macro attention
    shifts across bull -> base -> bear]",

```

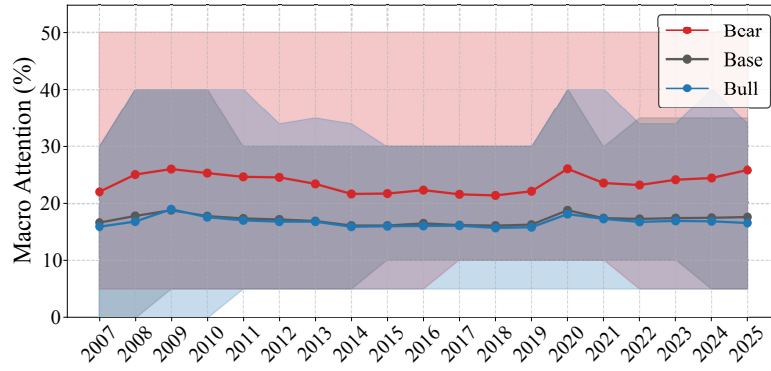
```
    "channel_evolution": "[summary of how dominant economic channels shift  
        across bull -> base -> bear]"  
  }  
}
```

Rules:

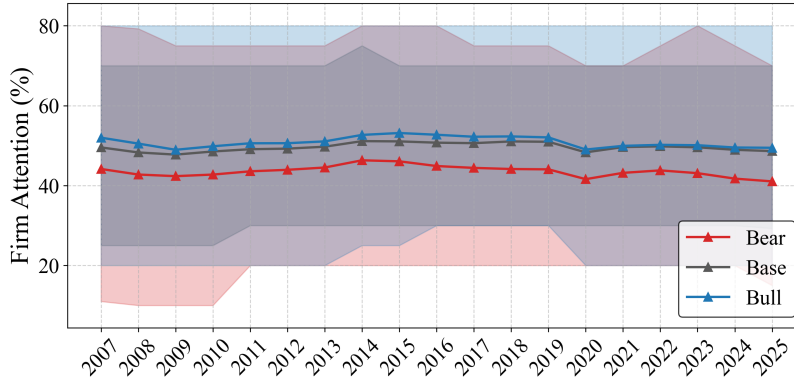
- All percentages must be integers (no decimals).
- Maintain an analytical and academic tone throughout.
- Do NOT include any text outside the JSON structure.
- Ensure structural consistency and valid JSON formatting.

B Additional Exhibits

Panel A: Macro Attention



Panel B: Firm Attention



Panel C: Industry Attention

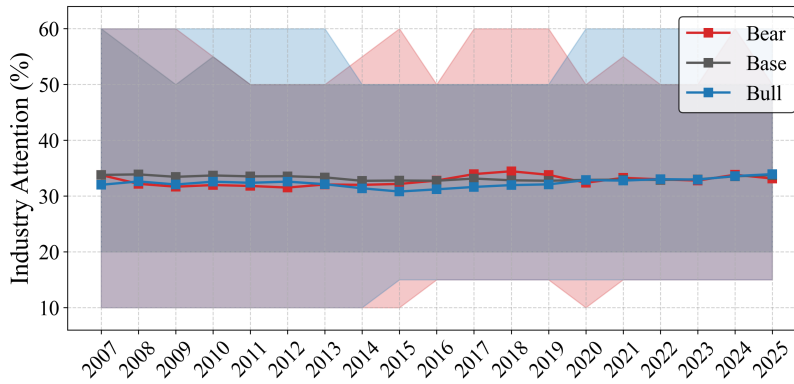


Figure A.1 Time Series of Attention to Firm, Industry, and Macro News.

Note: Time series of annual average attention shares to firm-, industry-, and macro-level news. The solid lines represent yearly means. The shaded areas correspond to the empirical 5th to 95th percentile range of the cross-sectional distribution in each year.

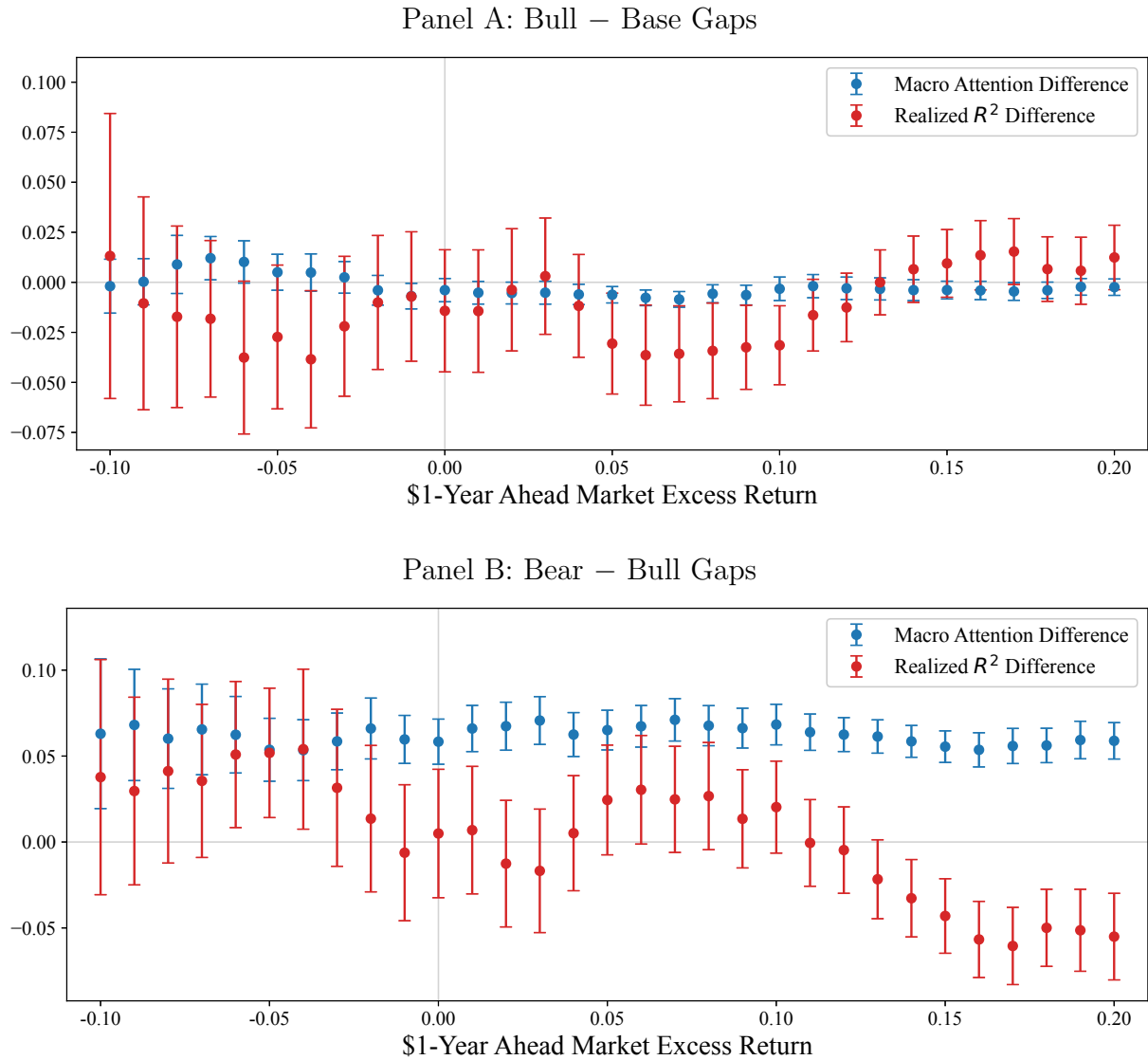


Figure A.2 Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bull–base and bear–bull differences in macro attention and realized CAPM R^2 across subsamples defined by 10% bins of forward 252-day market excess returns following the analyst report release. Each point includes reports with $x - 5\% \leq (R_m - R_f) \leq x + 5\%$ and applies the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bull, base, and bear cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bull, realized base, and realized bear cases. The plotted values represent the bull-minus-base and bear-minus-bull differences. Error bars show 95% confidence intervals from constant-only regressions.

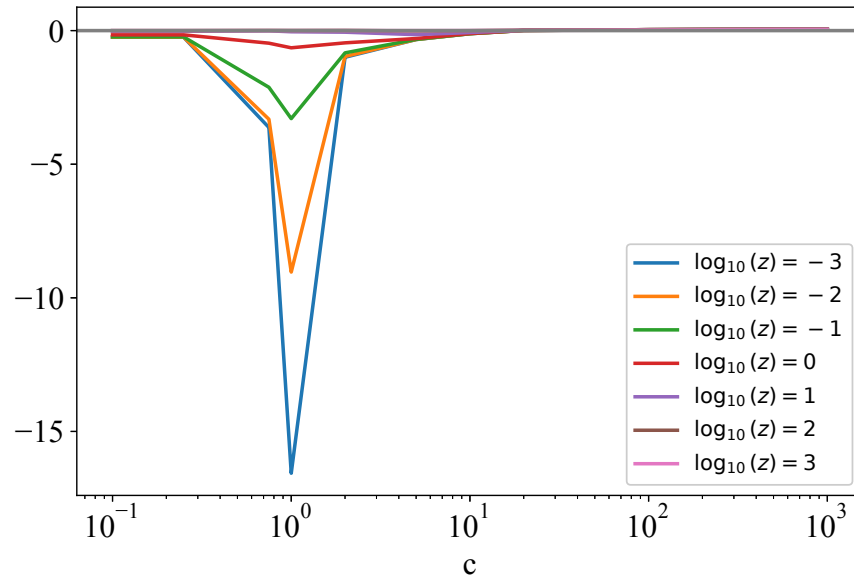


Figure A.3 Out-of-sample Market Timing Performance.

Note: This figure shows the out-of-sample R^2 that use RFFs to predict one-year ahead forecast error. The training window is $T = 12$ reports and RFF count P (or cT) ranges from 3 to 12,000 with $\gamma = 2$. The horizontal axis is the complexity level $c = P/T$.